INT396 - Unsupervised Learning | Section K3P24WV | CA-1 Groups and Projects

Find your group by your name or registration number. The objectives are what your project will be assessed on; the advanced extension is optional. Use the official source to read the original problem statement (for Smart India Hackathon problems, search the PS ID on the linked page).

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G1	Abhiram Alagandula (12406172) Sai Manasa Gade (12404684)	**Learning Compact Representations of Subsurface Ocean State**	Ocean interior state must be reconstructed from surface satellite observations, and the raw measurement space is far larger than the structure it contains.	Ocean research institutes, marine forecasting services	1. Build a study that learns a compact representation of ocean state and measures how much structure genuinely underlies the measurements. 2. Produce a reduced representation with variance explained and an intrinsic dimension estimate. 3. Report variance explained, reconstruction error, neighbourhood preservation. Baseline: random projection at equal dimension, mandatory. Deliver Ocean state explorer with 2D projection.	Dimensionality reduction on multivariate ocean fields; intrinsic dimensionality estimation	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Copernicus Marine and Argo float profiles (public) (Find and cite suitable public data yourselves.)	Compare linear and manifold reduction on structure preservation.	SMART INDIA HACKATHON 2026 PS ID: SIH26066 OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations. Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G2	Parise Mythily (12417095) Dushali Someshwar Ramteke (12416462)	**Structure Preservation in Super-Resolved Satellite Imagery**	Medium-resolution imagery is upsampled for mapping and land-cover work, and whether the added detail preserves real structure is rarely tested.	Remote sensing agencies, land-use analysts	1. Build a study that reduces imagery to a low-dimensional representation and measures how much genuine structure survives resolution change. 2. Produce a representation with measured neighbourhood preservation before and after resolution change. 3. Report neighbourhood preservation, reconstruction error, variance explained. Baseline: raw pixel projection, mandatory. Deliver Imagery explorer with projected structure view.	Dimensionality reduction on image patches; neighbourhood preservation measurement	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Sentinel-2 and Landsat imagery (public) (Find and cite suitable public data yourselves.)	Report the land-cover classes whose structure degrades most.	SMART INDIA HACKATHON 2026 PS ID: SIH26142 Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G3	L Teja Sai (12409306) Gagandeep Singh (12401857)	**Scale and Illumination Invariant Representations of Lunar Imagery**	Chandrayaan optical images of the same terrain differ with sun angle and scale, and correspondence must survive both.	Planetary science teams, mission operations	1. Build a study that learns an image representation and measures how much of its structure survives sun angle and scale change. 2. Produce a representation with measured correspondence stability under illumination and scale change. 3. Report neighbourhood preservation under transformation, reconstruction error, matching stability. Baseline: raw intensity correlation, mandatory. Deliver Correspondence explorer with matched pairs.	Representation learning; dimensionality reduction; invariance measurement	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Chandrayaan-2 OHRC and public lunar imagery archives (Find and cite suitable public data yourselves.)	Report the conditions under which correspondence fails entirely.	SMART INDIA HACKATHON 2026 PS ID: SIH26166 Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS) Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G4	Angelica Dhadwal (12405689) Shraddha Sharma (12400891)	**Latent Structure Discovery in Radio Signal Recordings**	Recorded IQ and audio captures are parameterised by hand, and the number of genuinely independent signal characteristics is never established.	Signal analysis teams, spectrum monitoring units	1. Build a study that reduces signal recordings to their independent characteristics and reports how many there genuinely are. 2. Produce a reduced signal representation with variance explained and an intrinsic dimension estimate. 3. Report variance explained, reconstruction error, separation of known classes. Baseline: hand-engineered parameter set, mandatory. Deliver Signal explorer with projected space.	Signal feature extraction; dimensionality reduction; intrinsic dimension estimation	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	RadioML public modulation datasets (Find and cite suitable public data yourselves.)	Test whether the representation separates modulation types without using labels.	SMART INDIA HACKATHON 2026 PS ID: SIH26147 Automated model for analysis of .IQ and .wav files along with signal parameter extraction Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G5	Himanish Giri (12406162) Shivam Bharti (12407787)	**Compact Motion Representations for On-Board Activity Recognition**	On-board systems must recognise activity with tiny compute budgets, so the representation must be as small as possible without losing structure.	Space mission operations, embedded systems teams	1. Build a study that compresses motion sensor data and reports how far it can be reduced before activity structure is lost. 2. Produce a reduced representation with the accuracy-versus-dimension curve reported. 3. Report variance explained, neighbourhood preservation, structure retained per dimension. Baseline: raw sensor channels, mandatory. Deliver Analysis notebook with the trade-off curve.	Sensor feature extraction; dimensionality reduction; compression-quality trade-off	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	PAMAP2 and UCI HAR activity datasets (public) (Find and cite suitable public data yourselves.)	State the smallest representation that still preserves activity structure.	SMART INDIA HACKATHON 2026 PS ID: SIH26174 AI Human Activity Recognition for On-board BAS Experiments Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G6	Shiro Mani Tripathi (12416040) Aarya Kumar (12405591)	**Low-Bitrate Speech Representations for Indian Languages**	Speech must travel over very low bitrate links, so the representation must keep what matters and discard the rest.	Radio communication services, multilingual speech technology teams	1. Build a study that learns compact speech representations and measures what is lost at each level of compression. 2. Produce a compressed representation with reconstruction quality reported per bitrate level. 3. Report reconstruction error, variance explained, per-language degradation. Baseline: uniform spectral downsampling, mandatory. Deliver Demo comparing reconstructions at each bitrate.	Speech feature extraction; dimensionality reduction; reconstruction quality measurement	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Common Voice and IndicSUPERB corpora (public) (Find and cite suitable public data yourselves.)	Report whether some Indian languages degrade faster than others.	SMART INDIA HACKATHON 2026 PS ID: SIH26173 iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G7	Fabian George Saji (12416200) Hinjis Christopher (12412439)	**Minimal Audio Representations for Always-On Edge Listening**	An always-on wake-word device must decide from the smallest possible representation because power and memory are severely limited.	Embedded device makers, voice interface teams	1. Build a study that finds how small an audio representation can be before wake-word structure disappears. 2. Produce a reduced audio representation with the structure-versus-size curve reported. 3. Report variance explained, neighbourhood preservation, separation of wake from non-wake. Baseline: full MFCC feature set, mandatory. Deliver Analysis notebook plus an edge-size recommendation.	Audio feature extraction; dimensionality reduction; size-versus-structure analysis	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Google Speech Commands dataset (public) (Find and cite suitable public data yourselves.)	State the smallest representation usable on a named microcontroller budget.	SMART INDIA HACKATHON 2026 PS ID: SIH26172 Low Latency and Efficient Voice Activator for Edge Devices Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G8	Sanskriti Khatri (12416885) Kondaveeti Preethi (12402506)	**Compact Visual Representations for On-Device Page Understanding**	On-device browser agents must understand a page from a representation small enough to run locally, and nobody has established how small that can be.	Browser and assistant engineering teams	1. Build a study that reduces page screenshots to a compact representation and measures what structure survives. 2. Produce a reduced visual representation with neighbourhood preservation reported per dimension. 3. Report neighbourhood preservation, reconstruction error, variance explained. Baseline: downscaled raw pixels, mandatory. Deliver Explorer showing the projected page space.	Image representation learning; dimensionality reduction; structure measurement	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	RICO webpage screenshot and UI element datasets (public) (Find and cite suitable public data yourselves.)	Report which page element types lose structure first.	SMART INDIA HACKATHON 2026 PS ID: SIH26171 On-device Visual Perception for Light-weight Browser Agents Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G9	Amitesh Krishna (12407919) Anshul Gurjar (12410893)	**Adaptive Resolution Representations of Lidar Point Clouds**	Lidar returns far more points than a vehicle can process, and uniform downsampling throws away exactly the detail that matters.	Autonomous vehicle perception teams	1. Build a study that reduces point clouds adaptively and measures where structure is lost against uniform downsampling. 2. Produce a reduced representation with structure preservation reported by object distance and size. 3. Report neighbourhood preservation, reconstruction error, per-class structure retained. Baseline: uniform voxel downsampling, mandatory. Deliver Point cloud viewer comparing reductions.	Point cloud feature extraction; dimensionality reduction; structure-preservation measurement	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	KITTI and nuScenes lidar data (public) (Find and cite suitable public data yourselves.)	Report which object classes degrade first under reduction.	SMART INDIA HACKATHON 2026 PS ID: SIH26053 Adaptive Variable Resolution 2.5D Lidar Mapping for Dynamic Environment Perception Organisation: DRDO https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G10	Vansh Ohlan (12406373) Sahil Kumar Gupta (12400926)	**Latent Structure in Inertial Navigation Drift**	Dead reckoning drifts in ways that look random but are driven by a small number of underlying factors nobody has enumerated.	Navigation system developers, fleet operators	1. Build a study that reduces inertial sensor error series to their underlying factors and reports how many there are. 2. Produce a factor decomposition of drift with variance explained and interpretation of each factor. 3. Report variance explained, reconstruction error, cross-device factor stability. Baseline: single cumulative drift metric, mandatory. Deliver Analysis dashboard with drift factors.	Sensor error feature engineering; dimensionality reduction; factor interpretation	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public inertial navigation and IMU benchmark datasets (Find and cite suitable public data yourselves.)	Test whether the factors are stable across devices.	SMART INDIA HACKATHON 2026 PS ID: SIH26168 AI-ML based Intelligent Dead Reckoning system for seamless navigation Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G11	Mudit Somra (12514098) Rupanshu Chawla (12412805)	**Reducing High-Dimensional Biomedical Data for Early Disease Signals**	Early disease detection works on biomedical data with far more measured features than patients, where most methods fail outright.	Clinical research teams, diagnostic laboratories	1. Build a study that reduces very wide biomedical data and establishes how much of it can be discarded safely. 2. Produce a reduced representation comparing at least three retention criteria on downstream separation. 3. Report variance explained, reconstruction error, downstream class separation. Baseline: retaining all features, mandatory. Deliver Analysis dashboard with retention comparison.	Dimensionality reduction on wide data; component retention criteria comparison	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	TCGA gene expression and public clinical omics datasets (Find and cite suitable public data yourselves.)	Estimate the intrinsic dimensionality and compare it against the retained count.	SMART INDIA HACKATHON 2026 PS ID: SIH26139 Hybrid Quantum Machine Learning Platform for Early Disease Detection Organisation: Egreen Quanta https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G12	Vanshika Joshi (12408898)	**Concept Structure Discovery in Quantum Algorithm Learning Material**	Learners are given quantum computing material organised by textbook chapter rather than by how the concepts actually relate.	Educators, learning platform designers, students	1. Build a study that reduces learning material to a concept space and shows how the topics genuinely relate. 2. Produce a concept space with variance explained and a proposed learning order. 3. Report variance explained, neighbourhood preservation, order agreement with expert. Baseline: chapter sequence, mandatory. Deliver Concept map explorer.	Text representation; dimensionality reduction; concept space interpretation	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Open quantum computing course corpora and arXiv abstracts (Find and cite suitable public data yourselves.)	Compare the derived order against the published syllabus order.	SMART INDIA HACKATHON 2026 PS ID: SIH26140 AI-Based Interactive Quantum Algorithm Learning Platform Organisation: Egreen Quanta https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G13	Vanshika Tyagi (12416151)	**Reducing Multimodal Remote Sensing Data for Interactive Query**	Remote sensing solutions are built one task at a time because the raw multimodal data is too large to explore interactively.	Remote sensing analysts, disaster and agriculture monitoring teams	1. Build a study that builds a compact shared representation across modalities so scenes can be explored by similarity. 2. Produce a shared representation with retrieval quality and neighbourhood preservation reported. 3. Report neighbourhood preservation, retrieval precision at K, variance explained. Baseline: single-modality representation, mandatory. Deliver Scene explorer with similarity search.	Multimodal representation learning; dimensionality reduction; retrieval by similarity	Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Sentinel-1/2 paired imagery and BigEarthNet (public) (Find and cite suitable public data yourselves.)	Report which modality contributes most to retrieval quality.	SMART INDIA HACKATHON 2026 PS ID: SIH26167 SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G14	Dev Mohit Bhardwaj (12407499) Pranshul Thakur (12411567)	**Service Co-Usage Discovery Across Government Digital Platforms**	Citizens move between disconnected government platforms and nobody knows which services are actually used together.	Digital governance bodies, service designers, citizens	1. Build a discovery system that mines which services co-occur in a citizen journey, to guide which platforms should integrate first. 2. Produce a ranked service co-usage rule set with support, confidence and lift. 3. Report support, confidence, lift; share of rules surviving lift filtering. Baseline: individual service popularity, mandatory. Deliver Integration planning dashboard.	Association rule mining over service usage baskets; lift-based filtering	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public e-governance service usage and transaction datasets (Find and cite suitable public data yourselves.)	Rank integration candidates by expected citizen benefit.	SMART INDIA HACKATHON 2026 PS ID: SIH26129 System integration and interoperability among government digital platforms,resulting in fragmented service delivery Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G15	Govind Sharma (12404801) Boosala Ujwal (12407989)	**Activity Sequence Mining in Infrastructure Project Monitoring**	Project monitoring records what happened but never which sequences of activity reliably precede a delay.	Project management units, contractors, funding agencies	1. Build a discovery system that mines the recurring activity sequences that precede schedule slippage. 2. Produce a ranked set of delay-preceding sequences with support and lift toward slippage. 3. Report support, confidence, lift toward delay; warning lead time. Baseline: milestone-slip counting, mandatory. Deliver Project dashboard with warning sequences.	Sequential pattern mining over project event logs; outcome-conditioned lift	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	BPI Challenge project and construction event logs (public) (Find and cite suitable public data yourselves.)	Report how early in a project the warning sequences appear.	SMART INDIA HACKATHON 2026 PS ID: SIH26103 Use case on web-based integrated project-monitoring platform Organisation: MoSPI https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G16	Abhyudai Sood (12400588) Aditya Kumar Singh (12404154)	**Query Sequence Mining for Multilingual Governance Assistance**	Citizens ask governance and legal questions in sequences that reveal what they are really trying to do, which single-question analysis misses.	Cooperative governance bodies, legal aid services	1. Build a discovery system that mines recurring query sequences to identify the underlying citizen tasks. 2. Produce a ranked set of recurring query sequences with the inferred task for each. 3. Report support, confidence, lift; task-completion rate per sequence. Baseline: single-query frequency, mandatory. Deliver Service design dashboard with task flows.	Sequential pattern mining over query sessions; task inference from sequences	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public query log and conversational assistance datasets (Find and cite suitable public data yourselves.)	Report which tasks are never completed within a session.	SMART INDIA HACKATHON 2026 PS ID: SIH26088 Multilingual Cooperative Governance & Legal Assistance Chatbot Organisation: Ministry of Cooperation https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G17	Sushil Kumar (12415141) Barun Chauhan (12404021)	**Craft Product Association Discovery for Artisan Cataloguing**	Artisan products are catalogued by craft type although buyers purchase across types in combinations nobody has measured.	Artisan cooperatives, marketing platforms, marginalised craftspeople	1. Build a discovery system that mines which craft products sell together, to guide cataloguing and bundling. 2. Produce a ranked product association rule set with lift separating genuine pairing from popularity. 3. Report support, confidence, lift; share surviving lift filtering. Baseline: raw co-occurrence counts, mandatory. Deliver Catalogue dashboard with bundle suggestions.	Association rule mining over purchase baskets; lift-based filtering	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public handicraft and e-commerce transaction datasets (Find and cite suitable public data yourselves.)	Translate the strongest rules into concrete bundle recommendations.	SMART INDIA HACKATHON 2026 PS ID: SIH26090 AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans Organisation: Ministry of Social Justice and Empowerment (MoSJE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G18	Ayush Dey (12404547) Tarun (12411401)	**Supply Chain Sequence Mining in Honey Traceability Records**	Traceability systems record every handoff but nobody mines which handling sequences precede a quality failure.	Beekeeping cooperatives, food safety regulators, buyers	1. Build a discovery system that mines the handling sequences associated with downstream quality problems. 2. Produce a ranked set of handling sequences with support and lift toward quality failure. 3. Report support, confidence, lift toward failure; earliest predictive handoff. Baseline: per-stage failure rates, mandatory. Deliver Traceability dashboard with risk sequences.	Sequential pattern mining over traceability events; outcome-conditioned lift	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public food supply chain and traceability event datasets (Find and cite suitable public data yourselves.)	Identify the earliest handoff at which the outcome becomes predictable.	SMART INDIA HACKATHON 2026 PS ID: SIH26021 Honey Chain: A block chain-based system for honey traceability and smart beekeeping management. Organisation: Ministry of MSME https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G19	Simran Choudhary (12411088) Kaajal Srii R (12405071)	**Learning Path Sequence Mining for Capacity Building**	Capacity building platforms offer courses in catalogue order, while the sequences that actually lead to completion are never mined.	Training institutions, learners, programme administrators	1. Build a discovery system that mines the course sequences leading to completion rather than dropout. 2. Produce a ranked set of course sequences with support and lift toward completion. 3. Report support, confidence, lift toward completion; dropout association. Baseline: course popularity ordering, mandatory. Deliver Learning path dashboard.	Sequential pattern mining over enrolment histories; outcome-conditioned lift	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Open University Learning Analytics Dataset (public) (Find and cite suitable public data yourselves.)	Report the sequences most strongly associated with dropout.	SMART INDIA HACKATHON 2026 PS ID: SIH26075 Participants are invited to design and develop **CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal** to support organizational training, competency development, and knowledge sharing through a centralized web-based platform. Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G20	Surendra Reddy Kota (12418513) Katikala Srisaa (12410329)	**Skill Co-Occurrence Mining for Cooperative Employment Planning**	Skills are trained one qualification at a time although employers hire for combinations no single qualification covers.	Cooperative employment programmes, training providers, jobseekers	1. Build a discovery system that mines which skills genuinely co-occur in demand, to guide combined training offers. 2. Produce a ranked skill association rule set with lift, and the training combinations it implies. 3. Report support, confidence, lift; share surviving lift filtering. Baseline: individual skill frequency, mandatory. Deliver Training planning dashboard.	Association rule mining over skill sets; lift-based filtering	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public job posting datasets with extracted skill lists (Find and cite suitable public data yourselves.)	Track how the strongest combinations shift over time.	SMART INDIA HACKATHON 2026 PS ID: SIH26087 AI-Enabled Cooperative Capacity Building, ERP & Employment Ecosystem Organisation: Ministry of Cooperation https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G21	Ommi Satya Pola Hemanth (12412388) N S Siddartha (12418846)	**Usage Sequence Mining for a Weather Application Homepage**	A weather application homepage is laid out by editorial judgement while user journeys follow sequences nobody has measured.	Meteorological service product teams, application users	1. Build a discovery system that mines recurring feature-usage sequences to guide what the homepage should surface. 2. Produce a ranked set of usage sequences with support and the homepage layout they imply. 3. Report support, confidence, lift; sequence coverage of sessions. Baseline: single-feature usage counts, mandatory. Deliver Product analytics dashboard with journey flows.	Sequential pattern mining over app event streams; support and lift	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public mobile application event log datasets (Find and cite suitable public data yourselves.)	Report whether sequences differ between monsoon and dry season.	SMART INDIA HACKATHON 2026 PS ID: SIH26076 Development of personalized homepage for 'Mausam' mobile application: Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G22	Nikunj Gujrati (12411132) Gaurav Singh Bhakuni (12404590)	**Protocol Deviation Sequence Mining in Clinical Trial Records**	Clinical trial monitoring reports deviations individually, so the recurring sequences that lead to them are never identified.	Clinical trial managers, regulatory monitors	1. Build a discovery system that mines the event sequences preceding protocol deviation. 2. Produce a ranked set of deviation-preceding sequences with support and lift. 3. Report support, confidence, lift toward deviation; warning lead time. Baseline: per-site deviation rates, mandatory. Deliver Trial monitoring dashboard with risk sequences.	Sequential pattern mining over trial event logs; outcome-conditioned lift	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public clinical trial monitoring and adverse event datasets (Find and cite suitable public data yourselves.)	Report how much warning the earliest sequence provides.	SMART INDIA HACKATHON 2026 PS ID: SIH26046 AIIA Clinical Trials Dashboard - a real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research, with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharma covigilance tracking. Organisation: Ministry of Ayush https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G23	kushwanth Gaddam (12405013) Krishanpal Singh (12417126)	**Medical Waste Composition Association Mining for Segregation Design**	Medical waste is segregated by rule although the categories that actually appear together vary by department in ways nobody has measured.	Hospital waste managers, municipal collection services	1. Build a discovery system that mines which waste categories co-occur by source, to redesign segregation at the point of generation. 2. Produce a ranked co-occurrence rule set with the segregation redesign it implies. 3. Report support, confidence, lift; projected contamination reduction. Baseline: prescribed category rules, mandatory. Deliver Waste management dashboard.	Association rule mining over waste manifests; lift-based filtering	Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public medical and municipal waste composition datasets (Find and cite suitable public data yourselves.)	Estimate the contamination avoided by acting on the top rules.	SMART INDIA HACKATHON 2026 PS ID: SIH26115 Design and Develop a Smart Mobile Medical-Waste Collection and Segregation System Organisation: Autodesk https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G24	Aryan Dixit (12419885) Monish Kumar Dewangan (12412915)	**Employment Outcome Pathway Grouping After Skilling**	Training systems capture enrolment and certification but lose track of what happens next, so outcomes are argued rather than measured.	Skilling missions, training providers, trainees	1. Build a system that groups trainees by post-training outcome pathway to show which pathways actually occur. 2. Produce an outcome pathway clustering with prevalence and profile per pathway. 3. Report silhouette, pathway separation, stability under attrition. Baseline: employed-versus-not binary split, mandatory. Deliver Outcome tracking dashboard.	Trajectory clustering on longitudinal outcome records; missing-data handling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Periodic Labour Force Survey and public skilling outcome datasets (Find and cite suitable public data yourselves.)	Report which trainee characteristics predict the weakest pathway.	SMART INDIA HACKATHON 2026 PS ID: SIH26135 Difficulties in tracking employment outcomes,skill gaps, and the impact of skilling initiatives Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G25	Jashandeep Singh Gill (12504818)	**Skill Demand Structure Discovery Against Training Provision**	Skill programmes are designed on historical occupation categories that no longer describe what employers are asking for.	Skill development councils, training institutions	1. Build a system that groups job demand by required skill signature and compares the groups against existing course categories. 2. Produce a demand clustering with disagreement against official course categories quantified. 3. Report silhouette, adjusted Rand index against categories, unserved-group count. Baseline: official occupation categories, mandatory. Deliver Curriculum planning dashboard.	Text feature extraction; demand clustering; comparison against taxonomy	Unit III clustering; Unit IV dimensionality reduction; Unit V cluster validity and evaluation	Public job posting datasets and NSQF course catalogue (Find and cite suitable public data yourselves.)	Report the demand groups no existing course serves.	SMART INDIA HACKATHON 2026 PS ID: SIH26134 Challenges in aligning skill development programs with industry requirements and emerging job market demands Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G26	Vaikhari Sanjay Matte (12411343) Ishu Pal (12408454)	**Archive Structure Discovery for a Digital Heritage Collection**	Heritage manuscripts and speeches sit across fragmented sources with no organisation, so they cannot be browsed or studied at scale.	Archives, historians, digital humanities researchers	1. Build a system that clusters archive documents into navigable themes and names each theme automatically. 2. Produce a document clustering with automatic labels and human-rated theme coherence. 3. Report topic coherence, human-rated label quality, silhouette. Baseline: chronological ordering, mandatory. Deliver Archive browser organised by discovered theme.	Text embedding; document clustering; automatic cluster labelling	Unit III clustering; Unit IV dimensionality reduction; Unit V cluster validity and evaluation	Public digitised heritage text collections (Find and cite suitable public data yourselves.)	Track how themes shift across the chronology of the collection.	SMART INDIA HACKATHON 2026 PS ID: SIH26096 Digital Heritage Archive for Memorials, Manuscripts & Ambedkar: AI-Powered Institutional Archive and Audio-Visual Knowledge Platform Organisation: Ministry of Social Justice and Empowerment (MoSJE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G27	Aditya Pal (12415931) Ankit Raj (12410782)	**Outreach Content Grouping for a Polar Science Repository**	Science outreach repositories grow without organisation, so material cannot be matched to the audience it suits.	Research outreach teams, educators, the public	1. Build a system that groups outreach material by content and reading level to match it to audiences. 2. Produce a content clustering with an audience profile per group. 3. Report silhouette, cluster coherence, audience coverage. Baseline: publication-date ordering, mandatory. Deliver Repository browser with audience filters.	Text embedding; clustering; readability feature engineering	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Open science outreach and educational text corpora (Find and cite suitable public data yourselves.)	Report which audiences the repository currently fails to serve.	SMART INDIA HACKATHON 2026 PS ID: SIH26063 Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G28	Manya Verma (12405212) Harleen Kaur Pannu (12416929)	**Alignment Drift Detection in Free Space Optical Camera Tracking**	Free space optical links fail when alignment drifts, and the drift is discovered only once the link has already dropped.	Optical communication operators, field engineering teams	1. Build a detector that learns normal alignment behaviour and flags drift before the link degrades. 2. Produce a drift detector with warning lead time before recorded link loss. 3. Report warning lead time, detection rate, false alarms per link-day. Baseline: fixed misalignment threshold, mandatory. Deliver Link health dashboard with drift trend.	Unsupervised drift detection on tracking series; healthy-baseline modelling	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public optical tracking and pointing-error datasets (Find and cite suitable public data yourselves.)	Separate genuine drift from transient atmospheric disturbance.	SMART INDIA HACKATHON 2026 PS ID: SIH26169 Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G29	Kaveri (12406280)	**Infrasound Event Discovery from High-Sensitivity Barometer Arrays**	Infrasound arrays record continuously and genuine events are buried in weather and cultural noise with no labels to learn from.	Geophysical monitoring agencies, disaster early warning services	1. Build a detector that learns the normal infrasound background and surfaces candidate events for analyst review. 2. Produce a candidate event ranking with precision at top-K against catalogued events. 3. Report precision at top-K, catalogued-event recall, false alarms per day. Baseline: amplitude thresholding, mandatory. Deliver Monitoring dashboard with candidate events.	Unsupervised anomaly detection on acoustic series; background modelling	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	IRIS and public infrasound array archives (Find and cite suitable public data yourselves.)	Cluster the candidates to separate event types from recurring noise sources.	SMART INDIA HACKATHON 2026 PS ID: SIH26144 Design & Development of a High-Sensitivity Micro barometer Infrasound sensor Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G30	Shubhank Bhatnagar (12403816) Siddha Tharun Sai (12410865)	**Movement Pattern Discovery for Rehabilitation and Ergonomics**	Rehabilitation and ergonomic assessment rely on clinician observation, so the movement patterns people actually use are never characterised at scale.	Rehabilitation services, occupational therapists, ergonomics teams	1. Build a system that groups recorded movements into patterns so therapy targets the pattern rather than a generic exercise. 2. Produce a movement pattern clustering with purity against annotated activity and per-person profiles. 3. Report cluster purity, adjusted Rand index, within-person consistency. Baseline: range-of-motion summary statistics, mandatory. Deliver Therapy dashboard with movement patterns.	Motion feature extraction; movement clustering; per-person profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	PAMAP2, UCI HAR and public rehabilitation motion datasets (Find and cite suitable public data yourselves.)	Detect when an individual pattern shifts, indicating progress or regression.	SMART INDIA HACKATHON 2026 PS ID: SIH26113 Human augmentation technologies are transforming healthcare,rehabilitation, industrial ergonomics, assistive living, sports, and personal mobility by improving human capabilities and enhancing quality of life. Organisation: Autodesk https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data